

Solution Architecture

Date: 29 September 2026 **Team ID:** SWTID-2026-4886
Project: AnswerBot – AI-powered FAQ and Customer Support Automation

Architecture Flow

Presentation / Client Layer	Postman / Thunder Client / Future Web or Mobile Client
API Layer	Express.js server (REST endpoints)
Security Layer	JWT authentication middleware + bcrypt
Core Services	Auth service FAQ service AI service Validation / Error handling
External Integration	Google Gemini API
Data Layer	MongoDB Atlas via Mongoose

Component Description

Component	Role	Technology
API Gateway	Receives HTTP requests, routing, JSON parsing and CORS	Express.js
Auth	Registration, login, token generation and protected routes	JWT, bcrypt
FAQ Service	CRUD, search and category operations	Node.js, Express, Mongoose
AI Service	Prompt handling and AI generation	Google Gemini SDK
Database	Users and FAQ documents	MongoDB Atlas
Error Handling	Consistent errors and safe responses	Express middleware